

vSphere

Deployment Guide

Study Notes with Real-World Examples

12

Chapters

100+

Concepts

2026

Edition

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2026 Edition

Table of Contents

1. What is Deployment in vSphere?	□
2. Installing ESXi — Step by Step	□
3. Deploying vCenter Server Appliance (VCSA)	□
4. vSphere Clusters — HA, DRS & EVC	□
5. Deploying Virtual Machines (VMs)	□
6. vSphere Networking for Deployment	□
7. vSphere Storage for Deployment	□
8. vSphere Lifecycle Manager (vLCM) — Patching & Upgrades	□
9. vSphere Security in Deployment	□
10. vSphere Fault Tolerance (FT) & vMotion	□
11. Backup & Disaster Recovery	□
12. Monitoring & Troubleshooting Deployments	□

Chapter 1

What is Deployment in vSphere?

Deployment in vSphere means setting up the virtual infrastructure — installing ESXi hypervisors on physical servers, deploying vCenter Server to manage them, and making sure everything is configured and ready to run virtual machines (VMs). Think of it like building the foundation of a house before you move furniture in.

The Big Picture — What Gets Deployed?

- ESXi Hypervisor

Installed directly on physical servers. It replaces the OS and turns bare metal into a VM host.

- vCenter Server Appliance (VCSA)

A Linux-based virtual appliance that gives you a single control panel to manage all ESXi hosts.

- Virtual Machines

The actual VMs — your Windows servers, Linux apps, databases — deployed on top of ESXi.

- Networking & Storage

Virtual switches, datastores, and storage adapters configured so VMs can communicate and store data.

Why Does Deployment Order Matter?

You cannot deploy vCenter before you have at least one ESXi host — because vCenter itself runs as a VM on ESXi! The correct order is: Install ESXi □ Deploy VCSA on ESXi □ Add more ESXi hosts to vCenter □ Configure clusters, networking, storage □ Deploy VMs.

□ Real-World Example

Real-World Example: Imagine you work at a bank. The bank wants to run 100 Windows servers without buying 100 physical computers. The IT team installs ESXi on 5 powerful physical servers, deploys vCenter to manage them, and then creates 100 VMs — all running on those 5 hosts. That is vSphere deployment in action.

□ Pro Tip

Always check the VMware Hardware Compatibility List (HCL) before installing ESXi on any server. Running on non-HCL hardware means VMware won't support you if something breaks.

Chapter 2

Installing ESXi — Step by Step

ESXi is a bare-metal hypervisor — it installs directly on the physical server with no underlying operating system. The installation is straightforward but knowing the requirements and options is key for the VCP-DCV exam.

Hardware Requirements for ESXi 8.0

- CPU

64-bit x86 processor with hardware virtualization support (Intel VT-x or AMD-V). Must have NX/XD bit enabled in BIOS/UEFI.

- RAM

Minimum 8 GB. Recommended 32 GB or more for production use.

- Storage / Boot Disk

Minimum 32 GB. SSD is strongly recommended. SD cards and USB drives are no longer supported for ESXi 8.0 boot.

- Network

At least one 1 Gbps NIC. 10 Gbps or faster recommended for vMotion and vSAN traffic.

- Firmware

UEFI firmware is recommended. BIOS is still supported but UEFI enables Secure Boot.

Three Ways to Install ESXi

- Interactive Install

Boot from ESXi ISO (USB or CD/DVD), follow the on-screen wizard. Best for small environments or first-time setup.

- Scripted Install (Kickstart)

Use a kickstart configuration file (ks.cfg) that automates all installation answers. Ideal for deploying 10+ hosts consistently.

- Auto Deploy (PXE Boot)

Hosts boot over the network and receive ESXi image + configuration from a vCenter Auto Deploy server. Stateless — no local ESXi installation needed. Best for large-scale environments.

What Happens During Installation?

The ESXi installer partitions the boot disk automatically — creating a VMFS-L (local) boot partition, a scratch partition for logs, and a VMFS datastore if space allows. After installation, the Direct Console User Interface (DCUI) appears on the physical monitor. From DCUI you set the management IP address so you can access the host remotely.

Post-Installation First Steps

- Set Management IP

Use DCUI (F2 key) to assign a static IP, subnet mask, gateway, and DNS for the management network.

- Configure NTP

Accurate time is critical for Kerberos auth and log correlation. Set NTP servers immediately after install.

- Enable SSH (if needed)

Disabled by default. Enable temporarily for troubleshooting, then disable again. Use lockdown mode in production.

- Accept EULA & License

Apply your ESXi license key via vCenter after adding the host, or via the Host Client directly.

▢ Real-World Example

Real-World Example: A retail company is opening a new data center. They have 20 identical Dell PowerEdge servers. Instead of visiting each server physically, the sysadmin creates a kickstart file with the IP scheme, root password, and NTP settings. All 20 ESXi hosts install automatically overnight without anyone sitting in front of each server.

▢ Pro Tip

During the exam — remember that ESXi 8.0 does NOT support booting from USB/SD cards as the primary boot device anymore. Use SSD or M.2 disk.

Chapter 3

Deploying vCenter Server Appliance (VCSA)

vCenter Server is the brain of your vSphere environment. Without it, you can only manage one ESXi host at a time using the Host Client. With vCenter, you get a single control panel for all your hosts, clusters, VMs, networking, and storage. The modern vCenter comes as VCSA — a pre-built Linux appliance you simply deploy as a VM.

VCSA vs Old Windows vCenter

- VCSA (Current Standard)

Linux-based OVA appliance. Embedded PostgreSQL database. Smaller footprint. Easier to patch and backup. This is the ONLY option from vSphere 7.0 onwards.

- Windows vCenter (Legacy)

Previously vCenter could be installed on Windows Server. Deprecated since vSphere 7.0. No longer supported or available for new deployments.

VCSA Deployment Stages

- Stage 1 — Deploy OVA

Mount the VCSA ISO on your laptop/workstation. Run the GUI or CLI installer. The installer deploys the VCSA appliance as a VM on a target ESXi host. Takes about 5–10 minutes.

- Stage 2 — Set Up vCenter

After the OVA is deployed, Stage 2 configures the vCenter services inside the appliance — SSO domain (vsphere.local), admin password, NTP, SSH settings, CEIP. Takes 10–20 minutes.

- Completed

After Stage 2, vCenter is accessible at <https://vcenter-hostname/ui> — the vSphere Client (HTML5).

Key VCSA Configuration Choices

- Deployment Size

Tiny (up to 10 hosts / 100 VMs), Small (100 hosts / 1000 VMs), Medium (400 hosts / 4000 VMs), Large (1000 hosts / 10000 VMs), X-Large (2000 hosts / 35000 VMs). Choose based on your environment size.

- Storage Size

Default vs Large vs X-Large — affects how much historical performance data and events are stored.

- SSO Domain

Default: vsphere.local. The administrator account is administrator@vsphere.local. You can join an existing domain for Enhanced Linked Mode.

- FQDN

Always use a proper Fully Qualified Domain Name (FQDN) for vCenter — e.g., vcenter.company.com. Using IP address causes certificate issues.

Enhanced Linked Mode (ELM)

If you have multiple vCenter Servers, Enhanced Linked Mode links them together so you can manage all of them from a single vSphere Client login. All vCenter instances must be in the same SSO domain. ELM also enables cross-vCenter vMotion — moving a VM from one vCenter's host to another vCenter's host without downtime.

▣ Real-World Example

Real-World Example: A hospital IT team manages two data centers — one in Mumbai and one in Pune. Each has its own vCenter. By configuring Enhanced Linked Mode, the IT admin can log into vSphere Client once and see all VMs in both data centers. During maintenance, they can live-migrate VMs from Mumbai to Pune using cross-vCenter vMotion without shutting anything down.

▣ Pro Tip

VCSA Stage 2 cannot be resumed if interrupted. If it fails, you must redeploy from Stage 1. Always ensure stable network and do not close the browser during Stage 2.

Chapter 4

vSphere Clusters — HA, DRS & EVC

A cluster is a group of ESXi hosts that share resources and work together. When you put hosts in a cluster, vSphere features like High Availability (HA) and Distributed Resource Scheduler (DRS) kick in — automatically restarting VMs on failure and balancing workloads across hosts.

vSphere High Availability (HA)

HA automatically restarts VMs on another host if their original host crashes. It uses heartbeat monitoring — hosts check on each other every second via the management network and via heartbeat datastores. If a host stops responding, the HA master declares it failed and restarts its VMs on surviving hosts.

- Admission Control

HA reserves cluster capacity to guarantee enough resources for VM restarts. Example: reserve 25% CPU and memory so 1 host failure can always be tolerated.

- VM Restart Priority

Set per VM — High, Medium, Low, Disabled. Critical VMs (like domain controllers) restart first.

- Heartbeat Datastores

HA uses 2 shared datastores as backup heartbeat channels. If management network fails but datastore heartbeat works, the host is NOT declared failed — just isolated.

- VM Component Protection (VMCP)

Restarts VMs if their datastore becomes permanently unavailable (PDL) — not just host failures.

Distributed Resource Scheduler (DRS)

DRS monitors CPU and memory utilization across all hosts in the cluster. If one host is overloaded and another is idle, DRS moves VMs via vMotion to balance the load — automatically, with no downtime.

- Automation Levels

Manual (only recommends), Partially Automated (auto initial placement, manual vMotion), Fully Automated (does everything automatically).

- DRS Rules

VM-VM Affinity: keep two VMs on the same host (e.g., app server + database for performance).

VM-VM Anti-Affinity: keep two VMs apart (e.g., two web servers on different hosts for HA).

- DRS Score (vSphere 7+)

Each VM gets a happiness score 0-100%. DRS migrates VMs to improve unhappy VMs rather than just equalizing host usage.

Enhanced vMotion Compatibility (EVC)

EVC solves a common problem: you have old servers with Intel Haswell CPUs and new servers with Intel Sapphire Rapids CPUs. Without EVC, vMotion between them fails because the CPU feature sets are different. EVC masks the newer CPUs down to the lowest common generation — all VMs see the same CPU features and can migrate freely between any host in the cluster.

- EVC Mode

Set at cluster level. Once set, cannot be disabled while VMs are running. Choose the lowest CPU generation you have in the cluster.

- Intel vs AMD

EVC modes are separate for Intel and AMD. You cannot mix Intel and AMD hosts in the same EVC cluster.

Cluster Requirements Checklist

- Shared Storage

All hosts must access the same datastores (SAN/NAS/vSAN) so DRS can vMotion VMs between them.

- vMotion Network

Each host needs a VMkernel port tagged for vMotion on the same VLAN/subnet.

- Compatible CPUs

Same CPU family required for vMotion — or use EVC to bridge generations.

- Same vCenter

All hosts in a cluster must be managed by the same vCenter (or ELM group for cross-vCenter vMotion).

▣ Real-World Example

Real-World Example: An e-commerce company runs its website on 6 ESXi hosts in a DRS+HA cluster. On Black Friday, traffic spikes and the web server VMs need more CPU. DRS automatically migrates some other VMs away from the busy hosts — giving the web servers the resources they need. Later that night, one host's power supply fails. HA detects the failure and restarts all VMs from that host on the remaining 5 hosts — shoppers never notice.

▣ Pro Tip

DRS requires Enterprise Plus licensing (or VVF/VCF subscription). HA is available in Standard edition and above. Always enable both together for production clusters.

Chapter 5

Deploying Virtual Machines (VMs)

Creating and deploying VMs is the most common day-to-day task in vSphere. You can create a VM from scratch, clone an existing VM, deploy from a template, or import an OVF/OVA package. Each method suits different use cases.

Method 1 — Create VM from Scratch

You manually pick the CPU count, memory, disk size, network adapter, and guest OS type. Then attach an ISO file and install the OS yourself. This is slow and inconsistent — not ideal for deploying many VMs.

Method 2 — Clone a VM

Cloning creates an exact copy of an existing VM including the OS, applications, and settings. You can clone a running VM with no downtime. After cloning, run Sysprep (Windows) or a Linux equivalent to give the new VM a unique identity (hostname, SID, IP address).

- Full Clone

A completely independent copy of the VM. Has its own VMDK files. Takes more storage but performs independently.

- Linked Clone

Shares the base VMDK with the parent VM. Uses much less storage but depends on the parent. Used heavily in VDI (Virtual Desktop Infrastructure).

Method 3 — Deploy from Template (Best Practice)

A template is a golden master VM — a VM that has been configured, patched, and had VMware Tools installed, then converted to a template so it cannot be accidentally powered on or modified. You deploy new VMs from the template in seconds, and use Customization Specifications to automatically set hostname, IP address, domain join, and time zone.

- Convert VM to Template

Right-click VM ▢ Template ▢ Convert to Template. The VM disappears from the VM list and appears under Templates.

- Clone to Template

Creates a template while leaving the original VM intact. Better for production — the original VM keeps running.

- Guest Customization Spec

A saved set of OS settings applied during deployment — hostname, IP, DNS, domain join, license key. Windows uses Sysprep under the hood.

- Content Library

Store templates centrally in a Content Library and sync them to other vCenter instances. Ensures all sites use the same approved golden images.

Method 4 — Import OVF / OVA

OVF (Open Virtualization Format) and OVA (single-file archive of OVF) are portable VM packages. Vendors ship pre-built appliances as OVA files — e.g., vCenter itself, NSX Manager, Veeam proxy, pfSense firewall. Deploy via: File ▢ Deploy OVF Template in vSphere Client, or use `govc` / `PowerCLI` for scripted imports.

VM Hardware Version

Every VM has a hardware version that determines which virtual devices and features it can use. vSphere 8.0 = Hardware Version 21. Higher versions support more vCPUs, RAM, NVMe controllers, USB 3.1, etc. Upgrading hardware version is irreversible — test in non-production first.

VMware Tools — Must Install

VMware Tools is a package installed inside the guest OS. Without it: vMotion may fail, graceful shutdown does not work, guest IP addresses are not visible in vCenter, snapshots are not quiesced (risking data corruption), and performance is reduced. Always install VMware Tools immediately after deploying a VM.

▢ Real-World Example

Real-World Example: A school's IT team needs to deploy 50 identical Windows 10 VMs for a computer lab. They build one perfect VM — install Windows, Office, antivirus, and VMware Tools — then convert it to a template. Using a Customization Specification that auto-assigns IP addresses from DHCP, they deploy all 50 VMs in under 30 minutes. Each VM gets a unique hostname and SID automatically via Sysprep.

▢ Pro Tip

Never use a VM snapshot as a long-term backup! Snapshots grow over time and hurt performance. They are for short-term rollback (patch testing, before upgrades) only.

Chapter 6

vSphere Networking for Deployment

Networking is the backbone of any vSphere deployment. VMs need to communicate with each other, with physical servers, and with the internet. ESXi uses virtual switches to connect VMs to the physical network – these work just like physical Ethernet switches, but in software.

Standard Switch (vSS) vs Distributed Switch (vDS)

- vSphere Standard Switch (vSS)

Configured individually per ESXi host. Free with all editions. Good for small environments. No central management – you must configure each host separately.

- vSphere Distributed Switch (vDS)

Centrally managed from vCenter. Configure once, push to all hosts automatically. Required for advanced features: LACP, NIOC, Port Mirroring, Private VLANs. Requires Enterprise Plus or VVF license.

Key Networking Concepts

- Port Group

A named group of ports on a switch with shared settings (VLAN, security, load balancing). VMs connect to port groups, not directly to the switch.

- VMkernel Port (vmk)

A virtual NIC for the ESXi host itself – NOT for VMs. Used for management traffic, vMotion, vSAN, iSCSI, NFS, and Fault Tolerance logging. Each vmk gets its own IP address.

- Uplink / VMNIC

Physical NIC on the ESXi server connected to the virtual switch. Always use at least 2 uplinks per switch for redundancy.

- VLAN Tagging (VST)

Set a VLAN ID on a port group. ESXi tags outgoing VM traffic with that VLAN ID and strips tags on incoming traffic. VMs don't need to know about VLANs.

NIC Teaming – Redundancy & Load Balancing

Connecting multiple physical NICs to a virtual switch is called NIC Teaming. It provides redundancy (if one NIC fails, traffic continues on the other) and optionally load balancing. Load balancing policies:

- Route by Virtual Port (Default)

Each VM's traffic always exits through the same uplink. Simple and works with any physical switch.

- Route by IP Hash

Traffic is distributed based on source+destination IP. Requires EtherChannel (LACP) configured on the physical switch.

- Route by Physical NIC Load (vDS only)

vDS automatically moves VMs to less-loaded uplinks. Best for environments with varied traffic patterns.

VMkernel Ports for Deployment

During deployment you will configure multiple VMkernel ports on each host:

- Management (vmk0)

Created automatically during ESXi install. The IP address you use to connect to the host.

- vMotion

Dedicated port for live VM migrations. Use a separate VLAN and 10 GbE or faster uplink.

- vSAN

Required on every host participating in a vSAN cluster. Uses dedicated VLAN and high-speed NICs.

- iSCSI / NFS

For connecting to SAN or NAS storage over the network. Jumbo frames (MTU 9000) recommended.

▣ Real-World Example

Real-World Example: A software company has a 4-host ESXi cluster. They configure a vDS with 4 uplinks per host – 2 uplinks for VM traffic (port group: Production-VMs, VLAN 100) and 2 uplinks for management + vMotion. Network I/O Control (NIOC) is configured to guarantee vMotion gets 4 Gbps bandwidth during busy times so VMs never stall mid-migration. When one NIC fails on a host, traffic automatically shifts to the second NIC and no VM notices the change.

▣ Pro Tip

Always separate management, vMotion, vSAN, and VM traffic onto different VLANs and VMkernel ports. Mixing them on the same interface creates security risks and performance problems.

Chapter 7

vSphere Storage for Deployment

VMs need storage for their virtual disks (VMDKs). vSphere supports multiple storage types — local disks, SAN (FC/iSCSI), NAS (NFS), and hyper-converged vSAN. Choosing the right storage type affects performance, cost, and which features (like vMotion and HA) you can use.

Storage Types at a Glance

- Local Storage (VMFS)

Disks physically inside the ESXi server. Fast and cheap. But VMs cannot vMotion to other hosts — the disk is stuck on one server. Only for dev/test or standalone hosts.

- SAN — Fibre Channel (FC)

Dedicated high-speed storage network using FC HBAs and FC fabric switches. Highest performance, lowest latency. Expensive. Used in banks, hospitals, large enterprises.

- SAN — iSCSI

SCSI storage commands over regular TCP/IP network. Uses standard NICs. More affordable than FC. Performance is excellent on 10 GbE networks with jumbo frames.

- NAS — NFS

File-level storage over TCP/IP. Easy to manage, flexible capacity. NFS v4.1 adds Kerberos security and multipathing. Good for general VM storage.

- vSAN

Aggregates local SSDs/NVMe from all ESXi hosts into a shared datastore. No external storage array needed. Built-in redundancy and deduplication. Ideal for HCI deployments.

VMFS — The VM File System

VMFS (Virtual Machine File System) is VMware's clustered file system for storing VM files on block storage (SAN/iSCSI). Multiple ESXi hosts can read and write to the same VMFS datastore simultaneously — this is what enables vMotion, HA, and DRS. Current version: VMFS6 (supports automatic space reclamation and 4K native drives).

Virtual Disk Formats — Thin vs Thick

- Thin Provisioned

The VMDK starts small and grows as data is written. A 100 GB thin disk might only use 10 GB of real space initially. Saves storage space but can cause issues if the datastore fills up.

- Thick Lazy Zeroed (Default)

Full disk space is reserved immediately but blocks are zeroed on first write. Good balance of speed and space management.

- Thick Eager Zeroed (EVT)

Full space reserved AND pre-zeroed at creation. Slowest to create but best performance. Required for vSphere Fault Tolerance.

vSAN — Hyper-Converged Storage

vSAN pools the local disks from all ESXi hosts in a cluster into one shared datastore. You need at least 3 hosts for fault tolerance. vSAN uses Storage Policies to define how many host failures a VM can survive. Example: FTT=1 means the VM's data is mirrored across 2 hosts — if one host fails, data is still safe on the other.

- Hybrid Mode

One SSD (cache) + multiple HDDs (capacity) per disk group. SSD caches reads and buffers writes.

- All-Flash Mode

One NVMe/SSD (write cache) + more SSDs (capacity). Much faster. Supports deduplication and compression.

- vSAN ESA (vSphere 8)

New architecture — single pool of all-NVMe drives. No separate cache tier. Native RAID-6 with no performance penalty.

- Witness Node

For 2-node vSAN or stretched clusters, a lightweight witness VM on a third site provides tie-breaking votes to prevent split-brain.

Storage Policy-Based Management (SPBM)

Instead of manually placing VMs on specific datastores, SPBM lets you define a policy (e.g., 'needs 2 copies of data, NVMe performance tier') and vSphere automatically places the VM on storage that meets those requirements. vSAN is fully SPBM-driven.

▣ Real-World Example

Real-World Example: A startup uses 4 ESXi servers each with 2 NVMe drives for vSAN. When they deploy a new database VM, they assign the 'Production-DB' storage policy: FTT=1 (mirror across 2 hosts), All-Flash, no encryption. vSAN automatically mirrors the VMDK across two hosts. When one host goes down for maintenance, the database keeps running on the mirrored copy with no downtime. No separate SAN hardware was purchased.

▣ Pro Tip

For vSAN to tolerate 1 host failure (FTT=1), you need at least 3 hosts. For FTT=2, you need 5 hosts. For RAID-6 erasure coding, you need at least 6 hosts.

Chapter 8

vSphere Lifecycle Manager (vLCM) — Patching & Upgrades

Keeping ESXi hosts patched and up to date is critical for security and stability. vSphere Lifecycle Manager (vLCM) — which replaced the older VMware Update Manager (VUM) — is the built-in tool for managing ESXi patches, upgrades, and firmware updates, all from vCenter.

Two Modes of vLCM

- Baseline Mode (Legacy)

You create baselines — collections of patches or upgrades — and attach them to hosts or clusters. Then scan (check compliance) and remediate (apply patches). This is the older VUM-compatible approach.

- Image / Desired State Mode (Modern)

You define the exact ESXi image (version + patches + drivers + firmware) you want on every host in a cluster. vLCM enforces this as the desired state — if a host drifts from the image, vLCM flags it and can automatically remediate. This is the recommended approach for new deployments.

Patching Workflow (Step by Step)

- Step 1 — Check Compliance

Scan the cluster against the baseline or desired image. vLCM shows which hosts are compliant and which need patches.

- Step 2 — Stage Patches

Download patches to the host without applying them. Staging reduces the maintenance window time.

- Step 3 — Enter Maintenance Mode

Before patching, put the host in Maintenance Mode. DRS automatically migrates all running VMs to other hosts in the cluster. VMs keep running — zero downtime for users.

- Step 4 — Remediate

vLCM applies the patches and reboots the host. After reboot, host exits Maintenance Mode and DRS moves VMs back.

- Step 5 — Rolling Update

Repeat for each host one at a time. The cluster remains operational throughout.

Quick Boot — Faster Patching

vSphere 6.7+ supports Quick Boot on compatible hardware. Instead of a full physical server reboot (which includes POST, BIOS, hardware init — often 10–15 minutes), Quick Boot restarts only the ESXi hypervisor software (1–2 minutes). This dramatically cuts patch maintenance windows for large clusters.

Hardware Firmware Management (vLCM + HW Vendors)

In Desired State mode, vLCM can also manage server firmware — BIOS, NIC firmware, HBA firmware, storage controller firmware — through vendor plugins (Dell OMIVV, HPE OneView). This ensures the entire software stack from firmware to ESXi is at a known-good validated state.

▣ Real-World Example

Real-World Example: A company has a 10-host cluster running production workloads 24/7. A critical security patch is released. The admin stages the patch to all hosts (downloads in background, no interruption). Then at 11 PM, they start the rolling update: Host 1 goes into Maintenance Mode, DRS moves its VMs to other hosts, patch applies and reboots (Quick Boot: 90 seconds), host comes back, moves to Host 2. All 10 hosts are patched by 1 AM. No VM was shut down. Users never noticed.

▣ Pro Tip

Always take a VCSA backup (File-Based Backup via VAMI port 5480) and a VCSA snapshot before upgrading vCenter. If something goes wrong, you can restore to the pre-upgrade state.

Chapter 9

vSphere Security in Deployment

Security must be built into every deployment decision — not added as an afterthought. From ESXi host hardening to VM encryption and role-based access control, vSphere provides multiple security layers that work together.

Role-Based Access Control (RBAC)

RBAC in vCenter controls who can do what. It has three parts: Users/Groups (from Active Directory or vsphere.local SSO), Roles (a named set of privileges — e.g., VM Power User, Read-Only), and Permissions (assigning a Role to a User on a specific object). Permissions flow down the inventory hierarchy — a permission set on a Datacenter applies to all clusters, hosts, and VMs inside it.

- Principle of Least Privilege

Give users only the permissions they need — nothing more. A helpdesk user who only needs to restart VMs should NOT have full Administrator access.

- Built-in Roles

Administrator (full access), Read-Only (view only), No Access, VM Power User, VM User, Resource Pool Admin, Datastore Consumer.

- Custom Roles

Create roles with exactly the privileges you need. Example: 'Backup Operator' role with only snapshot and datastore read permissions.

ESXi Lockdown Mode

Lockdown Mode forces all ESXi management through vCenter Server — you cannot SSH directly to the host or use the Host Client to make changes. This reduces the attack surface significantly.

- Normal Lockdown

Direct access disabled except for accounts on the DCUI Access list. SSH and ESXi Shell disabled.

- Strict Lockdown

Even the DCUI (physical console) is disabled. Only vCenter can manage the host. Recovery only through vCenter — if vCenter is down, you have a problem.

VM Encryption

vSphere VM Encryption encrypts all VM files — VMDK, VMX, swap files — at the hypervisor level using AES-256. The guest OS does not know about it. Encryption keys are managed by an external KMS (Key Management Server) or the built-in Native Key Provider (NKP). Encrypted VMs can still use vMotion (Encrypted vMotion) and snapshots.

vSphere SSO and Authentication

vCenter uses Single Sign-On (SSO) for authentication. The default SSO domain is vsphere.local with administrator@vsphere.local as the admin account. In production, always add Active Directory as an identity source so users log in with their corporate AD credentials instead of local SSO accounts.

Certificate Management (VMCA)

vSphere Certificate Authority (VMCA) automatically issues SSL certificates to all vSphere components. By default VMCA is a self-signed root CA. For enterprise environments, configure VMCA as a Subordinate CA under your existing enterprise CA (e.g., Microsoft Active Directory Certificate Services) so all vSphere certificates chain to your trusted enterprise root.

▣ Real-World Example

Real-World Example: A financial services company deploys vSphere for PCI-DSS compliance. They configure: AD integration for SSO (no shared local accounts), Strict Lockdown Mode on all ESXi hosts, VM Encryption for all cardholder database VMs, custom RBAC roles (DBA team can only manage their DB VMs, Network team can only manage vDS), and VMCA as Subordinate CA under their enterprise PKI. Every admin action is logged and forwarded to their SIEM (Splunk) via syslog.

▣ Pro Tip

Never use administrator@vsphere.local for day-to-day administration. Create named AD user accounts with appropriate roles. Keep the SSO admin account locked away for break-glass emergency access only.

Chapter 10

vSphere Fault Tolerance (FT) & vMotion

vMotion and Fault Tolerance (FT) are two of vSphere's most powerful availability features. vMotion moves running VMs between hosts with zero downtime. FT creates a real-time shadow VM that instantly takes over if the primary host fails — with zero data loss.

vMotion — Live VM Migration

vMotion transfers a running VM from one ESXi host to another with no downtime. The VM's memory is copied over the vMotion network while the VM keeps running. In the final milliseconds, the VM is briefly stunned, the remaining memory pages are copied, and the VM resumes on the new host. Users experience no interruption.

- Requirements

Shared storage accessible from both hosts, compatible CPUs (or EVC), vMotion VMkernel port on both hosts, same datacenter in vCenter.

- Encrypted vMotion

Encrypts VM memory in transit. Set per-VM: Disabled, Opportunistic (auto when supported), Required. Protects against network eavesdropping.

- Cross-vCenter vMotion

In Enhanced Linked Mode, you can vMotion VMs between hosts managed by different vCenter instances — even across data centers.

Storage vMotion (svMotion)

Storage vMotion moves a running VM's disk files from one datastore to another with no downtime. While the VM runs, ESXi mirrors writes to both old and new datastores simultaneously, then switches over. Use cases: migrating from slow storage to fast NVMe, balancing datastore utilization, decommissioning old storage arrays.

Fault Tolerance (FT) — Zero Downtime, Zero Data Loss

FT creates a secondary (shadow) VM on a different host that runs in perfect lockstep with the primary VM. Every CPU instruction is replicated to the secondary over a dedicated FT Logging network. If the primary host fails, the secondary VM takes over instantly — no reboot, no data loss, sub-second failover.

- FT Requirements

Dedicated FT Logging NIC (10 GbE recommended), EZT (Eager Zeroed Thick) disk format, FT-compatible CPU, FT licensing.

- FT Limitations

Maximum 8 vCPUs per FT-protected VM. No snapshots, Storage vMotion, or linked clones on FT VMs.

- FT vs HA

HA restarts VMs after a host failure (minutes of downtime). FT provides zero downtime and zero data loss — but costs more resources (the shadow VM consumes as much CPU and RAM as the primary).

▣ Real-World Example

Real-World Example: A stock exchange runs its order-matching engine as a VM with Fault Tolerance enabled. Each trade takes microseconds — any downtime means financial loss. When a host hardware failure occurs, the FT secondary takes over in under a second. Traders never see an interruption. The failed host is replaced, and FT re-establishes the secondary VM on the repaired host automatically.

▣ Pro Tip

Use FT only for truly mission-critical VMs. FT doubles the resource consumption (CPU + RAM) of each protected VM and limits the VM to 8 vCPUs. For most VMs, HA provides sufficient protection at much lower cost.

Chapter 11

Backup & Disaster Recovery

Deployment is not complete without a backup and DR plan. vSphere provides the VADP (vStorage APIs for Data Protection) framework that lets backup tools protect VMs efficiently, and vSphere Replication for replicating VMs to a secondary site for disaster recovery.

VADP — How vSphere Backup Works

VADP lets backup software (Veeam, Commvault, Veritas) take VM backups without installing agents inside each VM. The backup software talks to vCenter, takes a VM snapshot, reads the VMDK data via the vSphere API, sends it to the backup repository, then deletes the snapshot. Changed Block Tracking (CBT) ensures only changed blocks are backed up after the first full backup — making incremental backups fast.

- Changed Block Tracking (CBT)

ESXi tracks which disk blocks have changed since the last backup. Incremental backups only copy changed blocks — reducing backup time from hours to minutes.

- Transport Modes

Hot-Add (proxy VM mounts VMDK directly — fast), SAN (proxy reads LUN directly — fastest), NBD (over network — universal but slower).

- Application-Consistent Backup

VMware Tools triggers VSS (Volume Shadow Copy) inside Windows VMs to flush application data to disk before the snapshot — ensuring database-consistent backups.

vSphere Replication — Site-Level DR

vSphere Replication (VR) asynchronously replicates VMs from a primary site to a recovery site over any IP network. RPO (Recovery Point Objective) is configurable from 5 minutes to 24 hours. No shared storage between sites is required — VR sends changed blocks over the network.

- RPO

How much data you can afford to lose. RPO=5 min means at most 5 minutes of changes are lost if the primary site goes down.

- RTO

How long it takes to get VMs running at the recovery site. With vSphere Replication + SRM, this can be minutes.

Site Recovery Manager (SRM)

SRM automates the entire failover process. You define Recovery Plans — scripted runbooks that specify the order VMs restart, IP address remapping, and any custom scripts to run. SRM supports: Non-disruptive DR testing (VMs start in an isolated bubble — production continues), Planned Migration (graceful move with zero data loss), and Emergency Failover (when primary site is gone).

VCSA Backup — Protect vCenter Itself

vCenter's own configuration and database must be backed up separately. The built-in File-Based Backup (FBB) at VAMI (<https://vcenter:5480>) backs up the entire VCSA to SCP/SFTP/NFS/SMB. Schedule daily backups. If vCenter is lost, deploy a fresh VCSA from ISO and restore from backup.

▣ Real-World Example

Real-World Example: A manufacturing company has its ERP system (SAP) running on vSphere in Mumbai. Using vSphere Replication with a 15-minute RPO, all SAP VMs are continuously replicated to their Pune DR site. SRM is configured with a Recovery Plan that starts the database VM first, then the application servers, then the web tier — with proper startup delays. During their annual DR drill, they test the failover in 12 minutes without impacting production. RPO achieved: 15 minutes. RTO achieved: 12 minutes.

▣ Pro Tip

A VM snapshot is NOT a backup. Snapshots live on the same datastore as the VM — if the datastore fails, you lose both the VM and the snapshot. Always use a proper backup solution with off-site storage.

Chapter 12

Monitoring & Troubleshooting Deployments

After deployment, you need to continuously monitor performance, quickly identify issues, and troubleshoot problems before they impact users. vSphere provides multiple built-in tools for monitoring, and understanding the key metrics is essential for the VCP-DCV exam.

Key Performance Metrics to Watch

- CPU Ready (%RDY)

Time a VM waited for a physical CPU. Above 5% means CPU contention on the host. Fix: reduce vCPU count, vMotion VMs to less loaded host, add hosts to cluster.

- Memory Balloon (MCTLSZ)

Amount of guest memory being reclaimed via the balloon driver. Non-zero means the host is under memory pressure. Fix: add RAM to host, set memory reservations, vMotion VMs.

- Disk Latency (DAVG)

Average storage response time per command. Above 20 ms for SAN or 25 ms for SATA indicates storage performance issues.

- Dropped Network Packets

Non-zero values mean NIC saturation or misconfiguration. Check uplink utilization and NIC teaming policy.

esxtop — Real-Time Performance on ESXi

esxtop is the ESXi equivalent of Linux 'top'. Access via SSH to the host and run: esxtop. Press 'c' for CPU, 'm' for memory, 'd' for disk, 'n' for network. Sort by pressing Shift+F and selecting a field. Use batch mode (esxtop -b -d 5 -n 10 > output.csv) to capture performance data over time for analysis.

Common Deployment Issues & Fixes

- Host shows 'Not Responding' in vCenter

Check management network (ping the host). SSH to host and restart vpxa service (service vpxa restart). Check vmkernel.log for errors.

- vMotion fails

Check: CPU compatibility (enable EVC), vMotion VMkernel port connectivity, shared storage accessibility, no DirectPath I/O or USB passthrough devices on VM.

- VM won't boot

Check VM events in vCenter, verify datastore accessibility, look for snapshot chain issues (Consolidation Needed warning), check vmware.log in VM's datastore folder.

- ESXi PSOD (Purple Screen)

Take a photo of the screen (shows error code), collect vm-support bundle, check hardware health in iLO/iDRAC. Open VMware support case.

- Datastore inaccessible

Rescan storage adapters, check SAN zoning/iSCSI connectivity, verify LUN still presented from storage array, check esxcli storage nmp device list for path status.

vSphere Health Checks

vSphere Client □ Cluster □ Monitor □ vSphere Health provides automated checks for: configuration best practices, vSAN health, certificate expiry, HCL compliance, and more. Run health checks regularly — especially before and after any major change.

□ Real-World Example

Real-World Example: After a weekend patch cycle, users report that some VMs on a particular host are slow. The admin SSHs to the host and runs esxtop. They see %RDY (CPU Ready) values of 25% on several VMs — far above the 5% threshold. Investigation reveals one newly patched VM is running a runaway process consuming 100% CPU. The admin kills the process in the guest OS, CPU Ready drops to 0%, and all VMs on the host return to normal performance.

□ Pro Tip

Set up vCenter Alarms for: host disconnection, datastore usage above 80%, CPU Ready above 10%, HA failover events. Proactive alerts save you from discovering problems through angry user calls.

These study notes cover the VMware VCP-DCV 2024 — Deploy topic with 12 chapters, real-world examples, and concept diagrams. Combine with hands-on lab practice on VMware Workstation or a home ESXi lab for best exam results.

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